

IGCSE • Cambridge (CIE) • Maths

 1 hour  14 questions

Target Test

Created 29/05/26

Total Marks

/63

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1

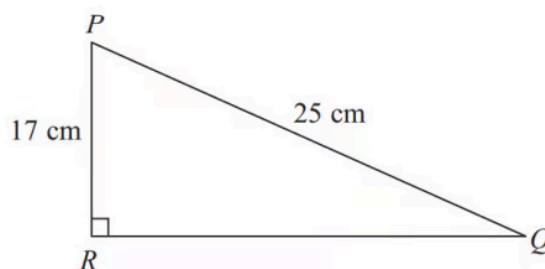


Diagram **NOT**
accurately drawn

PQR is a right-angled triangle.

$$PR = 17\text{ cm}$$

$$PQ = 25\text{ cm}$$

Work out the size of angle RPQ .

Give your answer correct to 1 decimal place.

(3 marks)

2

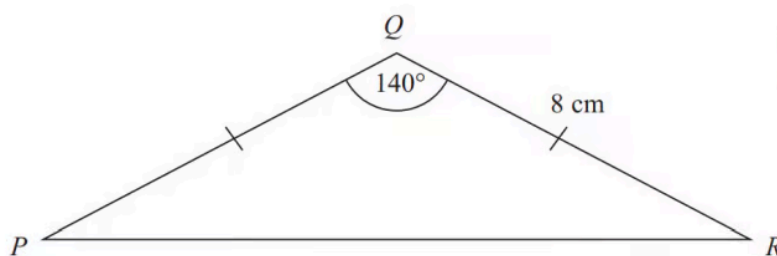


Diagram **NOT**
accurately drawn

Calculate the length of PR .

Give your answer correct to 3 significant figures.

(3 marks)

3 Here is a rectangle.

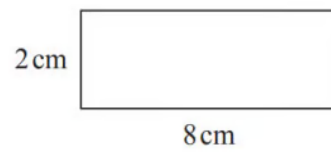


Diagram **NOT**
accurately drawn

The 8-sided shape below is made from 4 of these rectangles and 4 congruent right-angled triangles.

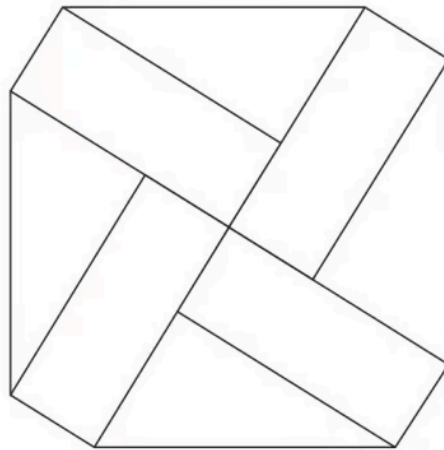


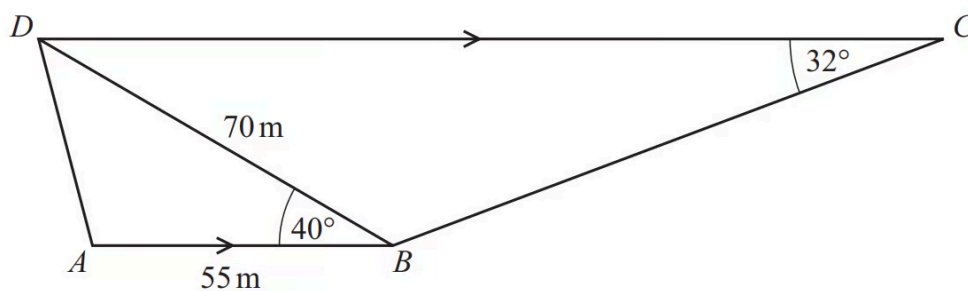
Diagram **NOT**
accurately drawn

Work out the perimeter of the 8-sided shape.

You must show all your working.

(5 marks)

4

NOT TO
SCALE

The diagram shows a trapezium $ABCD$.

AB is parallel to DC .

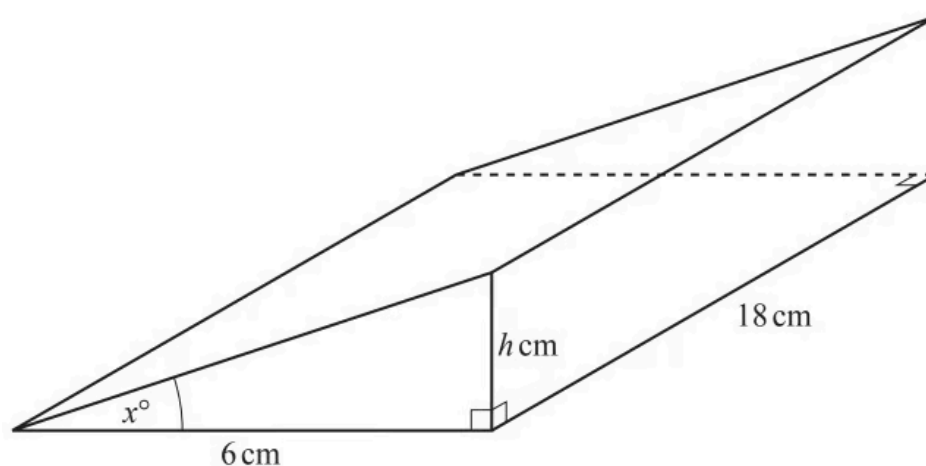
$AB = 55$ m, $BD = 70$ m, angle $ABD = 40^\circ$ and angle $BCD = 32^\circ$.

Calculate the shortest distance from A to BD .

..... m

(2 marks)

5 (a)

NOT TO
SCALE

The diagram shows a prism with length 18 cm and volume 253.8 cm^3 .

The cross-section of the prism is a right-angled triangle with base 6 cm and height h cm.

Show that the value of h is 4.7 .

(3 marks)

(b) Calculate the value of x .

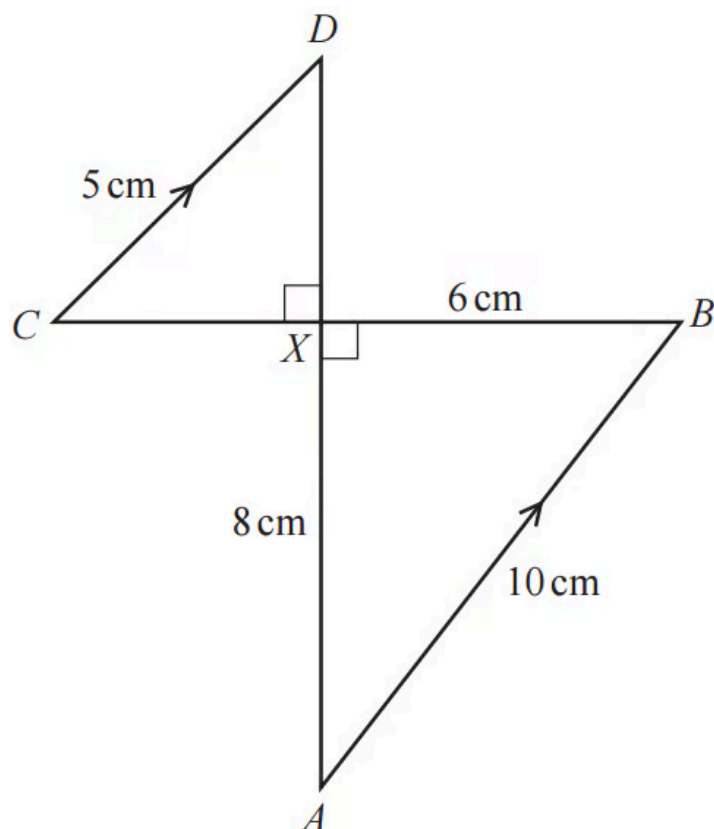
$x = \dots\dots\dots$

(2 marks)

(c) Calculate the total surface area of the prism.

$\dots\dots\dots \text{cm}^2$

(6 marks)



NOT TO
SCALE

In the diagram, AB and CD are parallel.
 AD and BC intersect at right angles at the point X .
 $AB = 10$ cm, $CD = 5$ cm, $AX = 8$ cm and $BX = 6$ cm.

Calculate angle XAB .

Angle $XAB = \dots\dots\dots$

(2 marks)

7 The diagram shows a rectangular framework.

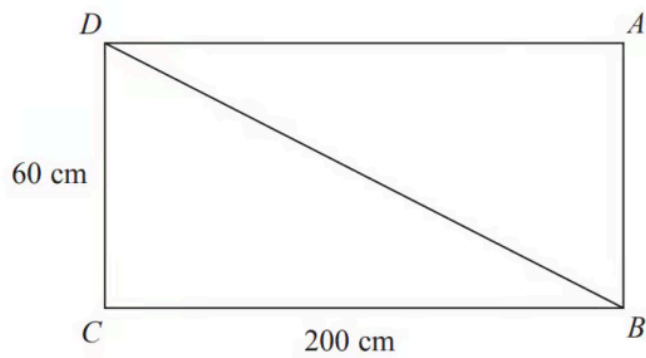


Diagram **NOT**
accurately drawn

The framework is made from 5 metal rods.

The metal rods have a weight of 0.9 kg per metre.

Work out the total weight of the framework.

Give your answer, in kg, correct to 3 significant figures.

(4 marks)

8

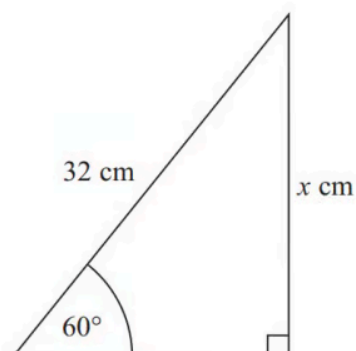


Diagram **NOT**
accurately drawn

Calculate the value of x .

Give your answer as an exact value.

(3 marks)

- 9 The diagram shows a rectangle and a diagonal of the rectangle.

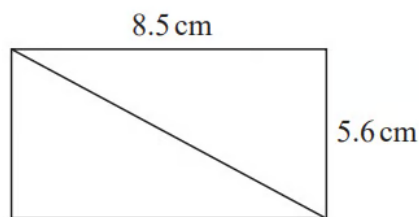


Diagram **NOT**
accurately drawn

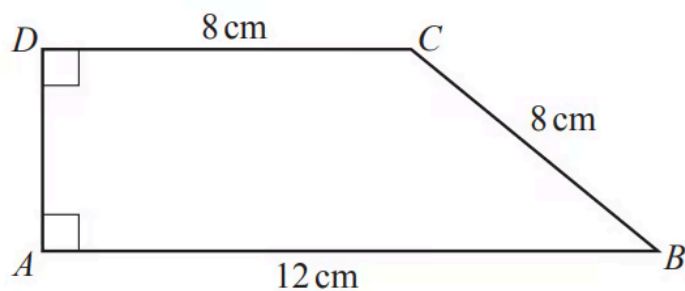
Work out the length of the diagonal of the rectangle.

Give your answer correct to 1 decimal place.

.....cm

(3 marks)

10 Calculate the area of this trapezium.



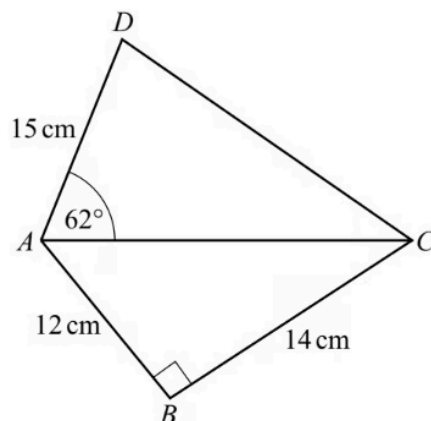
NOT TO
SCALE

Give your answer as an exact value.

..... cm^2

(4 marks)

11 (a)



NOT TO
SCALE

The diagram shows a quadrilateral, $ABCD$, formed from two triangles, ABC and ACD .

ABC is a right-angled triangle.

Calculate angle BAC .

Angle $BAC = \dots\dots\dots$

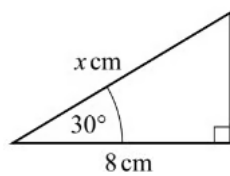
(2 marks)

(b) Calculate the shortest distance from D to AC .

$\dots\dots\dots$ cm

(3 marks)

12



NOT TO
SCALE

Find the exact value of x .

$x = \dots\dots\dots$

(4 marks)

13 Here is part of a field.

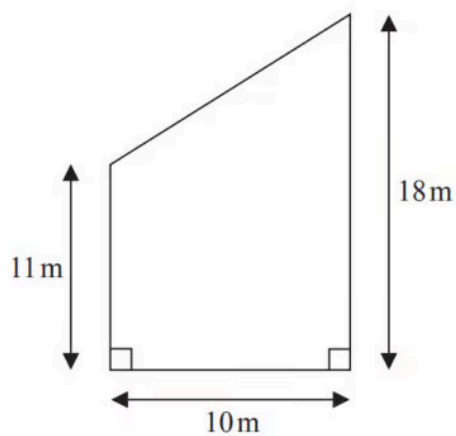


Diagram **NOT**
accurately drawn

This part of the field is in the shape of a trapezium.

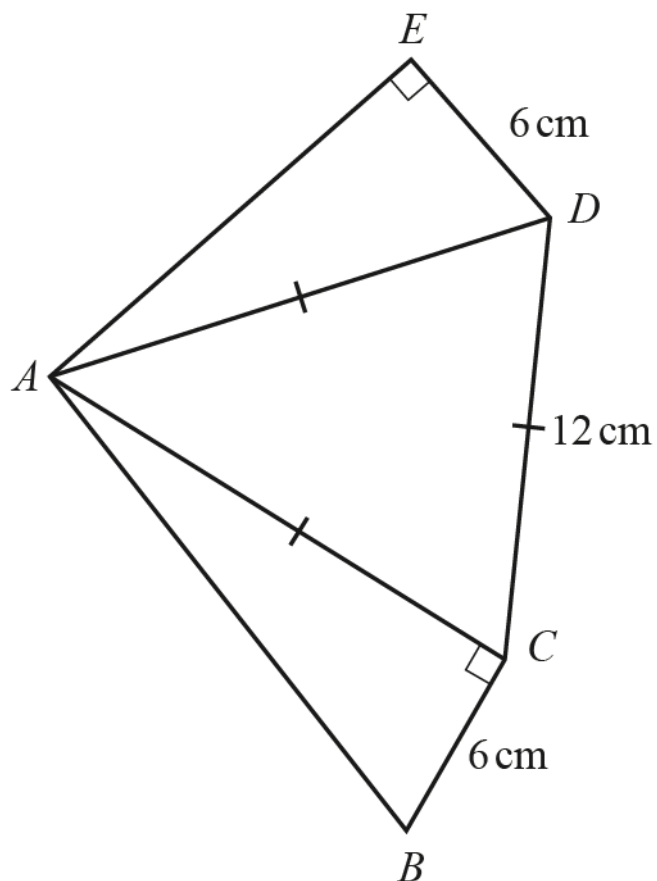
A farmer wants to put a fence all the way around the edge of this part of the field.

The farmer has 50m of fence.

Does he have enough fence? You must show all your working.

(5 marks)

14 (a)



NOT TO
SCALE

In the pentagon $ABCDE$, angle $ACB = \text{angle } AED = 90^\circ$. Triangle ACD is equilateral with side length 12 cm.

$DE = BC = 6 \text{ cm}$.

Calculate angle BAE .

Angle $BAE = \dots\dots\dots$

(4 marks)

(b) Calculate AB .

$$AB = \dots\dots\dots \text{ cm}$$

(2 marks)

(c) Calculate AE .

$$AE = \dots\dots\dots \text{ cm}$$

(3 marks)